Electronic Circuits

Powering Our World with Tinkercad

Grade 6 Science & ICT

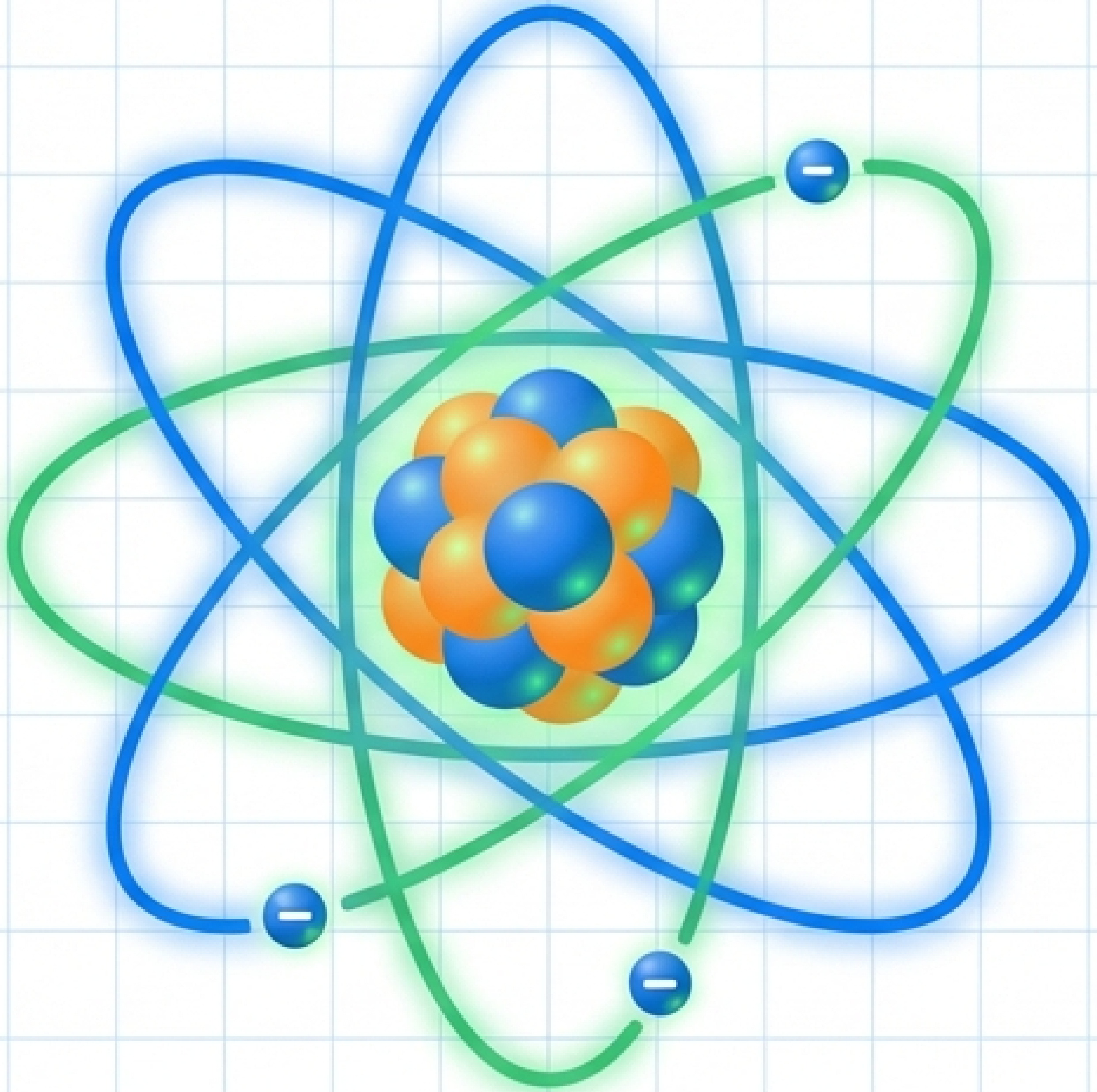
Explaining the invisible world of electricity.

NotebookLM

The Invisible World

Protons (+):

Positively charged particles in the center.



Electrons (-):

Negatively charged particles zooming around.

Electricity is the flow of Electrons moving through a device!

The Big Three Concepts

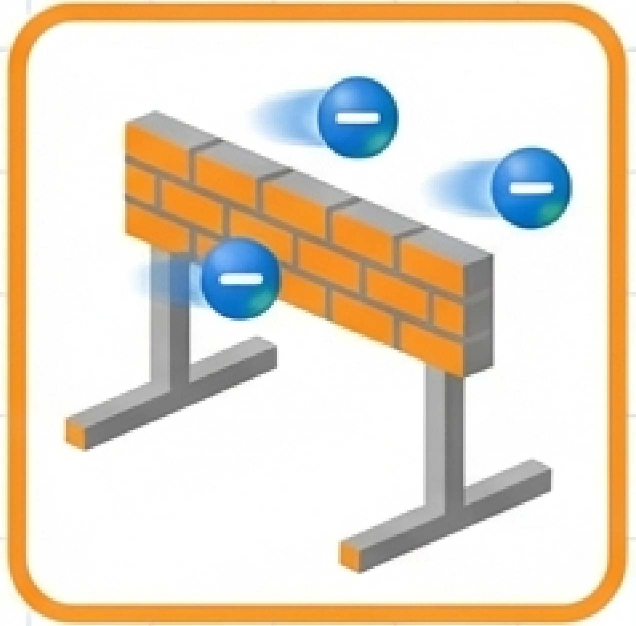
Current (I)

The flow of electrons. Measured in Amperes (A).

Voltage (V)

The 'push' or pressure moving the electrons. Measured in Volts (V).

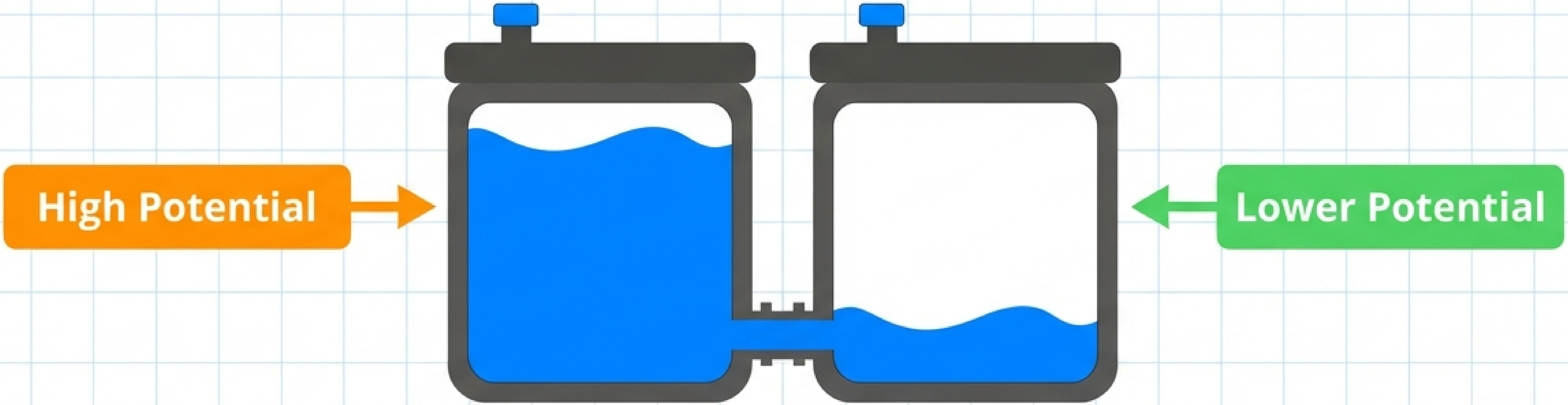
Resistance (R)



The force slowing down the flow. Measured in Ohms (Ω).

Did You Know? A battery is like a storage tank for this energy push!

Understanding Voltage: The Water Analogy



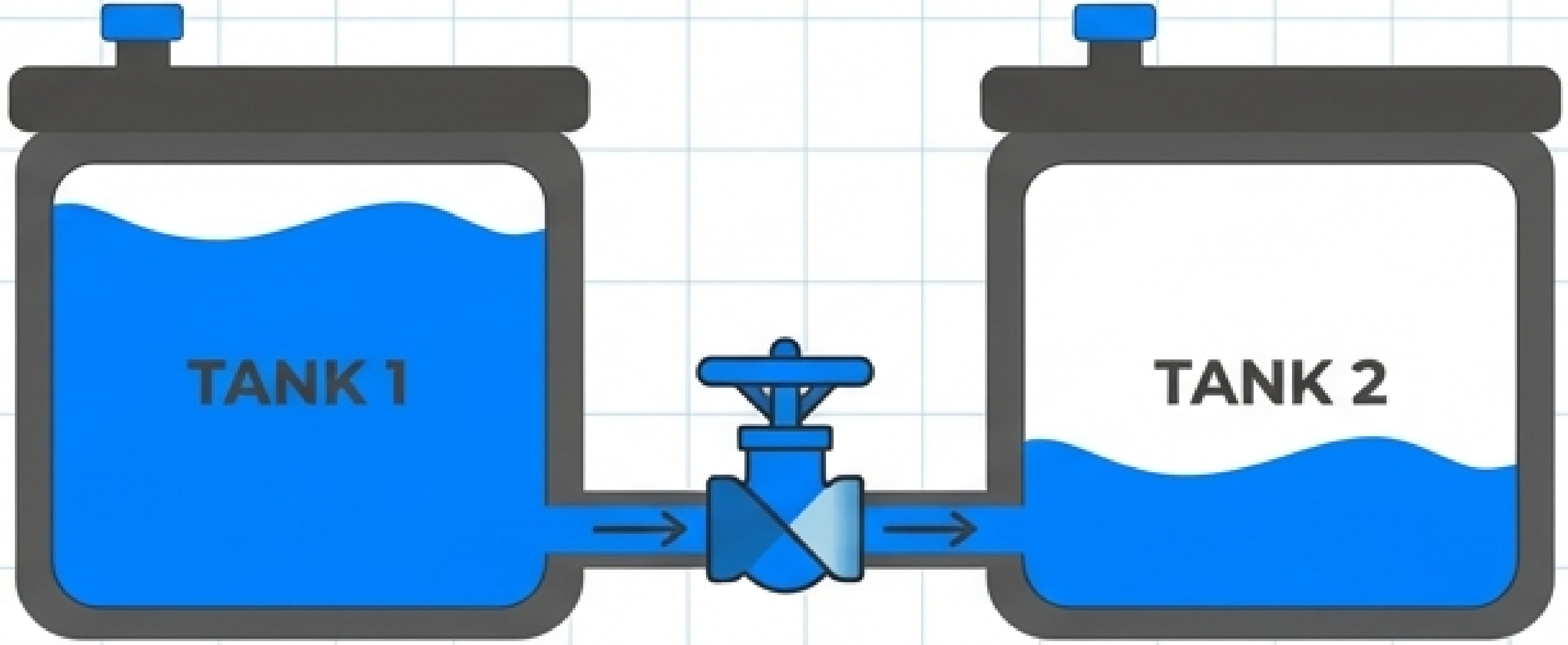
Just like water flows from High to Low levels,
Electricity flows from High to Low potential.

Water Pressure = Voltage (The Push)

Understanding Resistance: The Control Valve

- Uncontrolled flow is dangerous (like a **flood**).
- Resistance acts like a **valve** or **dam gate**.
- To Resist means **'To Hold'**.

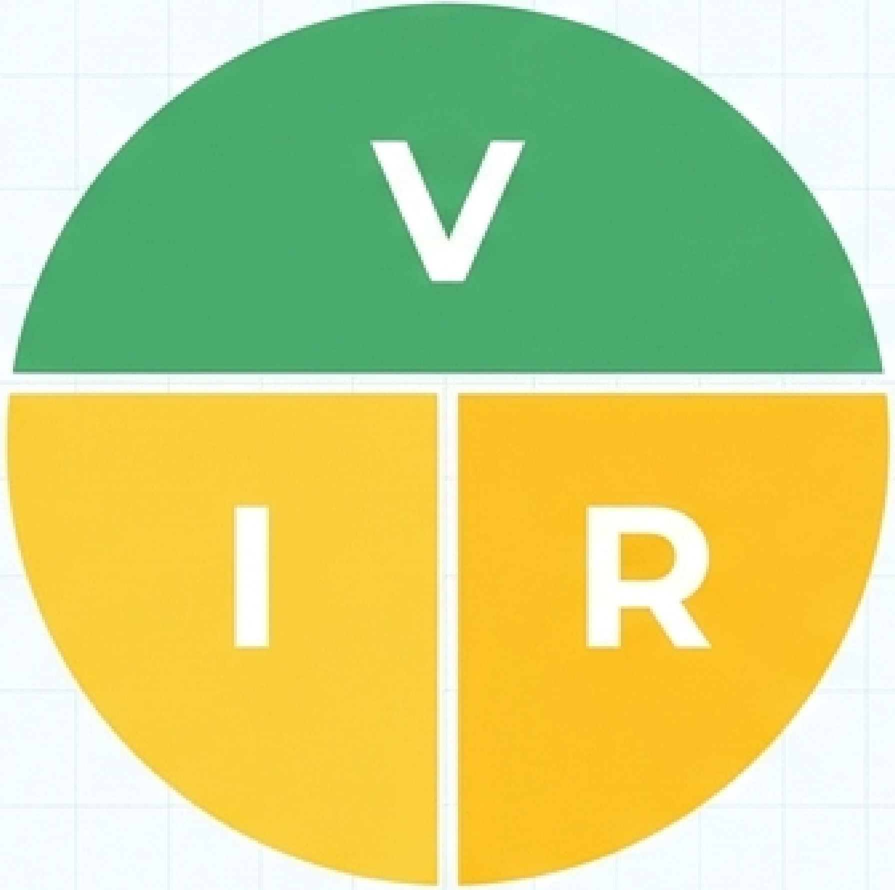
Uncontrolled flow is dangerous (like a flood).



Resistance acts like a valve or dam gate.

Higher Resistance = Lower Current Flow.

The Rules of the Road: Ohm's Law



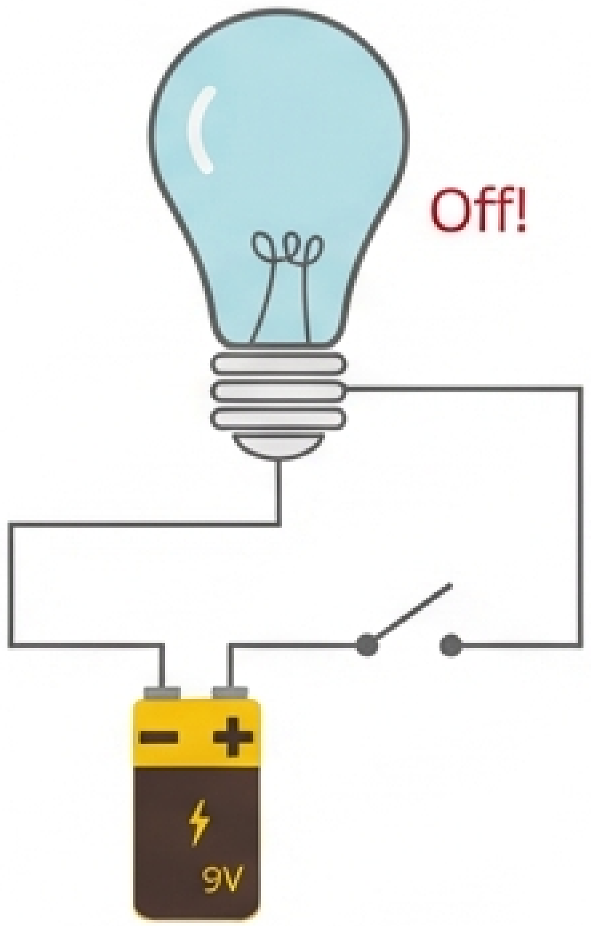
Current (I) = Voltage (V) ÷ Resistance (R)

If Voltage goes UP, Current goes UP. ↑

If Resistance goes UP, Current goes DOWN. ↓

The Path: Open vs. Closed Circuits

Open Circuit



- Path is broken. No flow.
- LED is OFF.

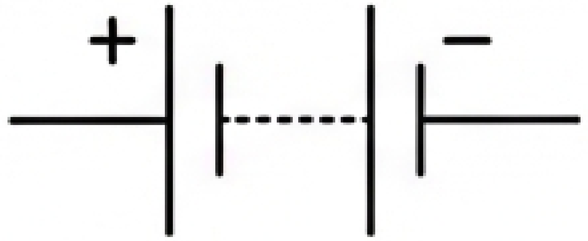
Closed Circuit

- Path is complete. Current flows.
- LED is ON!

Your Toolbox

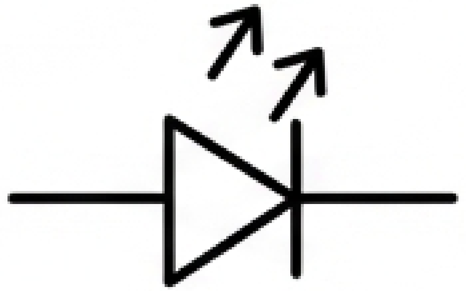
9V Battery

The Power Source.



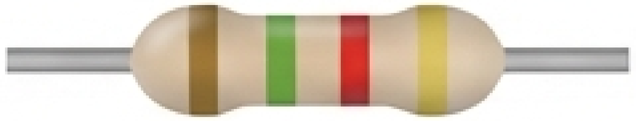
LED

Light Emitting Diode (The Output).



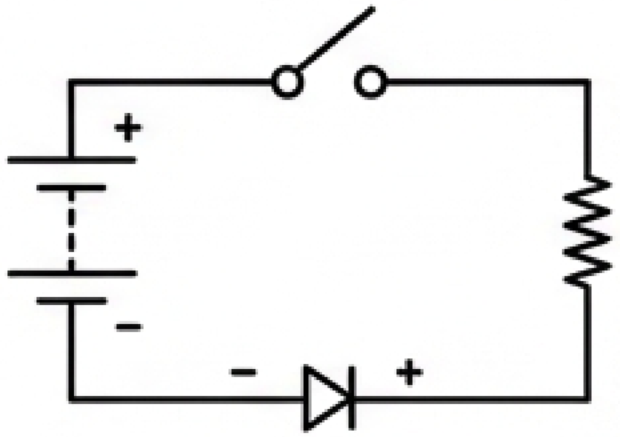
Resistor

Limits the current (Measured in Ohms).

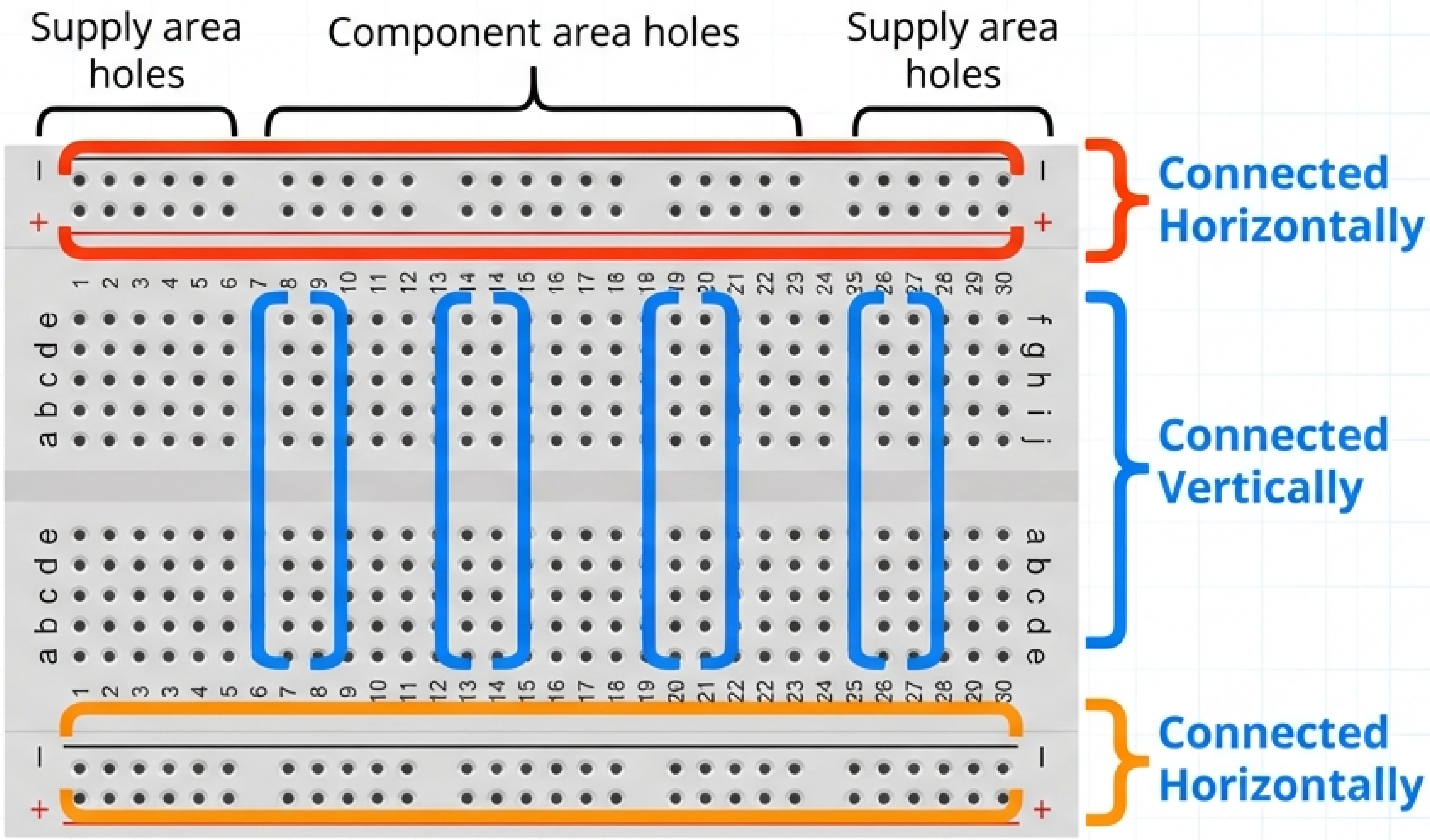


Pushbutton

The Switch.



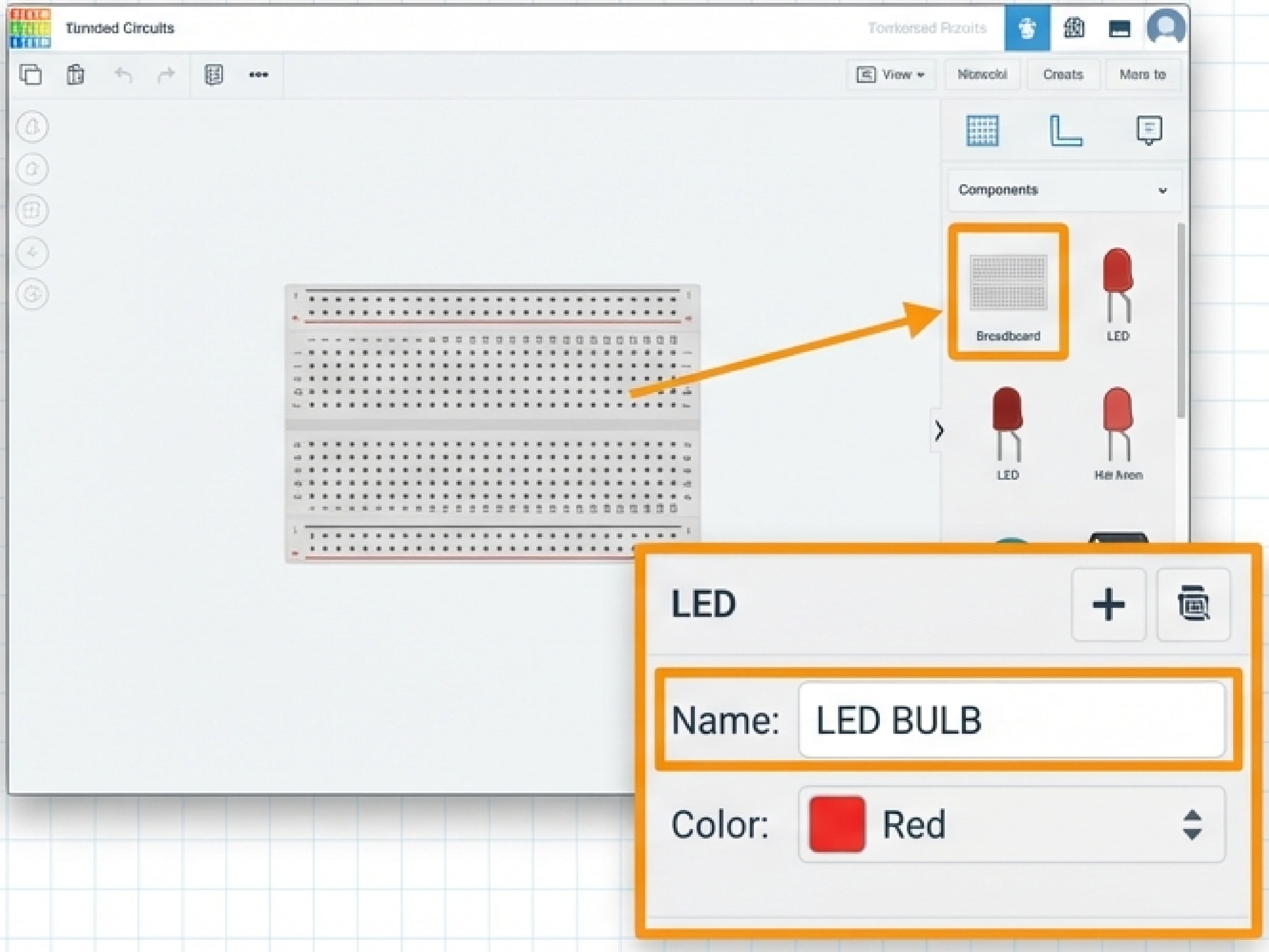
The Virtual Lab: How a Breadboard Works



Rule: Connect component legs into the same vertical column to join them!

Let’s Build! Step 1: Setup

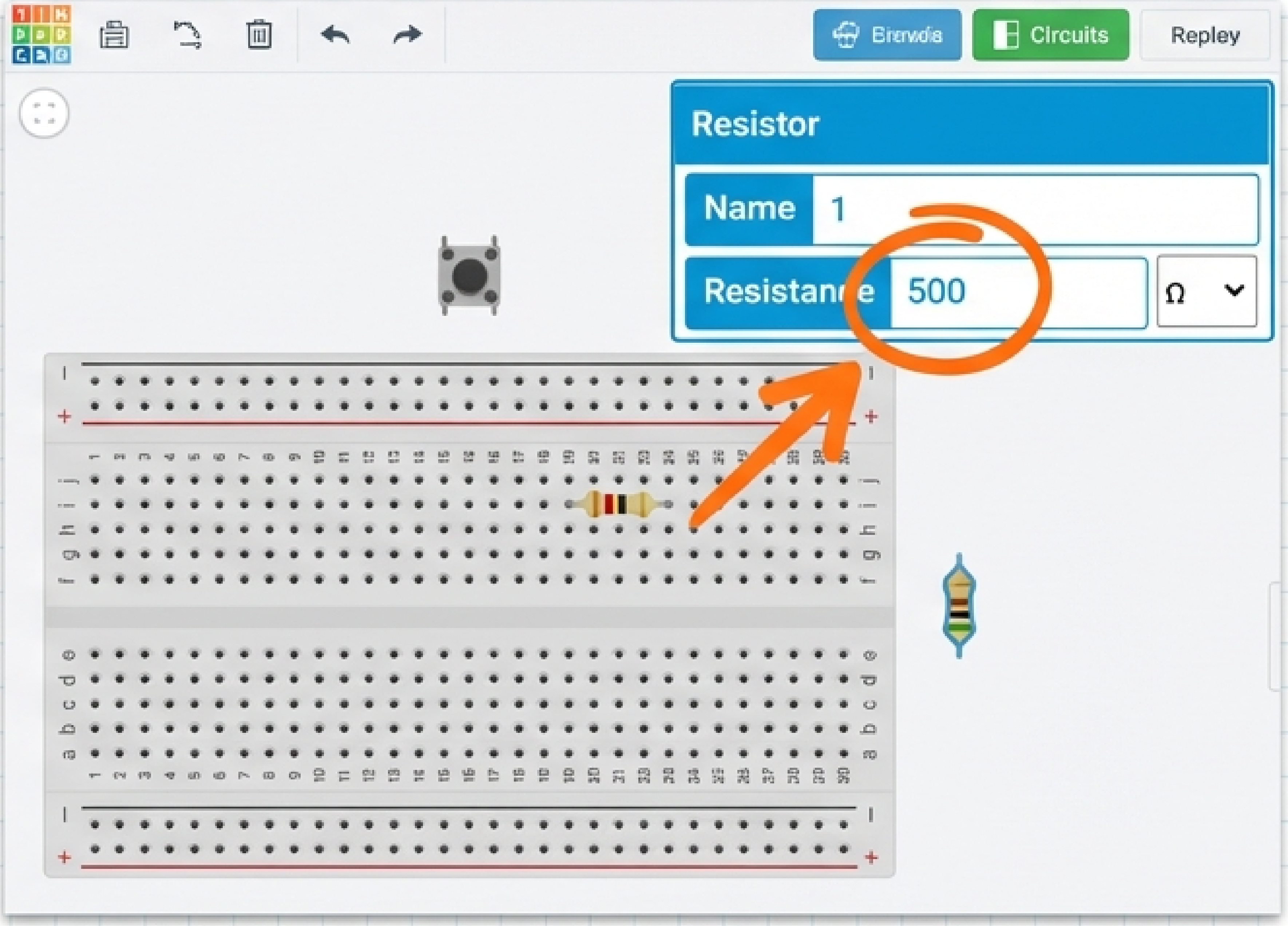
1. Log into Tinkercad Circuits.
2. Drag a Small Breadboard into the workspace.
3. Drag an LED onto the board.
4. Name your LED "LED BULB" and pick a color.



Let’s Build! Step 2: Components & Values

1. Add a Pushbutton (Name it “Switch”).
2. Add a Resistor. **IMPORTANT:** Change **Resistance** to 500 Ω (Ohms).
3. Add a 9V Battery and rotate it to face the board.

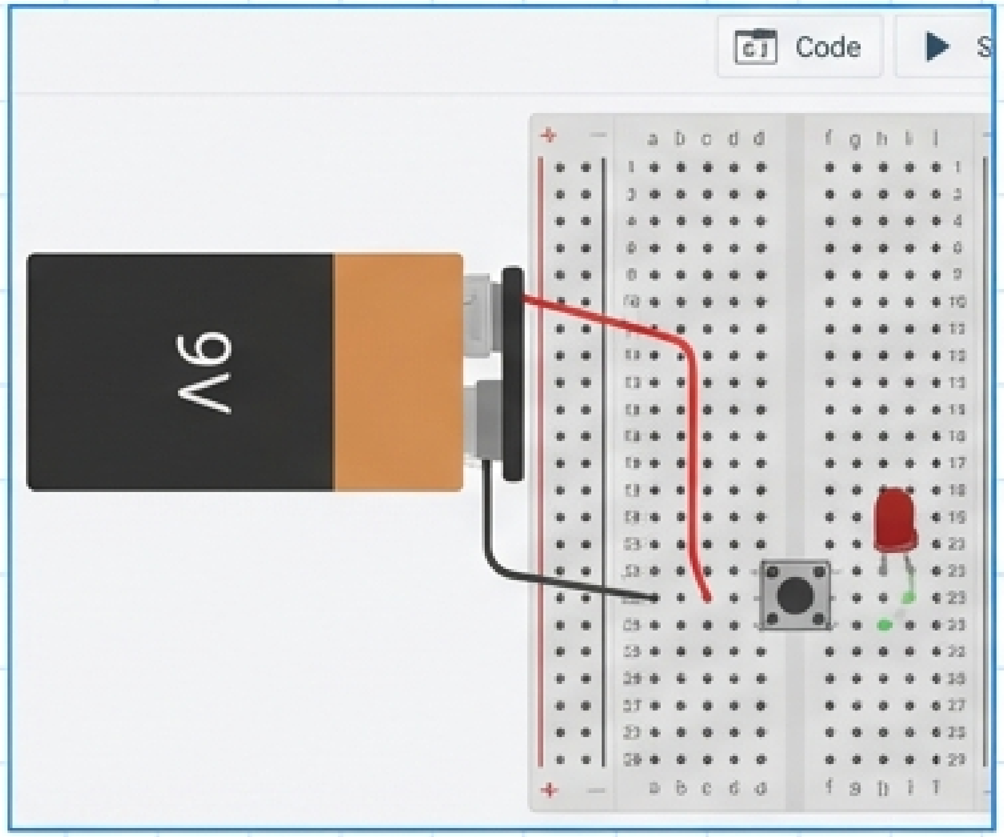
Note: We need 500Ω to protect the LED from the strong battery!

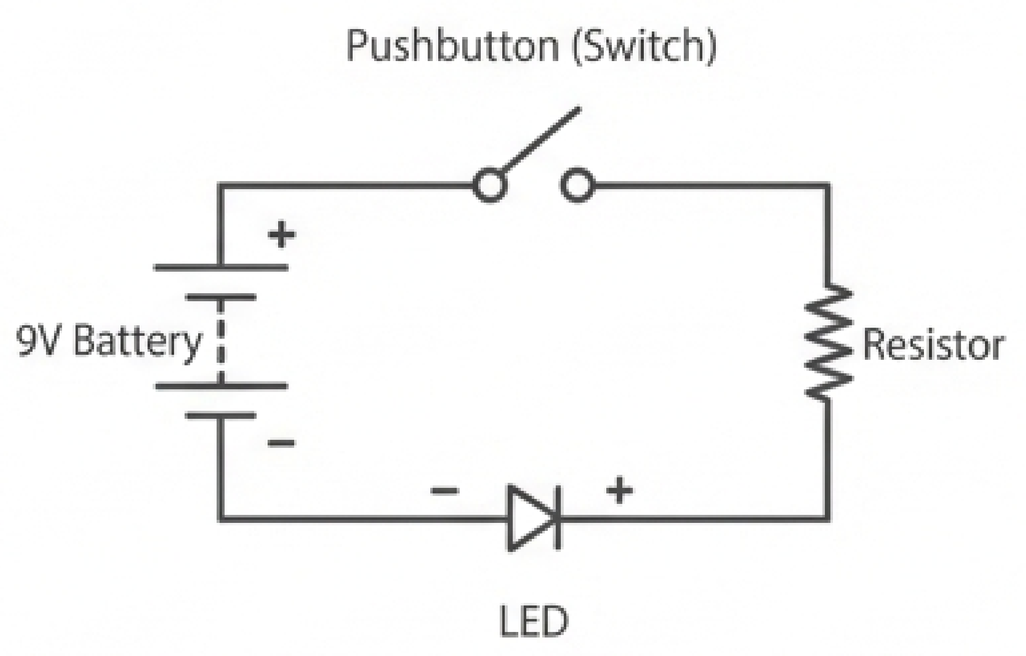


Let's Build! Step 3: Wiring the Path

Wiring Checklist

- **Red Wire:** Connect Battery Positive (+) to the Switch.
- **Black Wire:** Connect Battery Negative (-) to the LED cathode.
- **The Loop:** Ensure the Resistor connects the LED to the Switch.



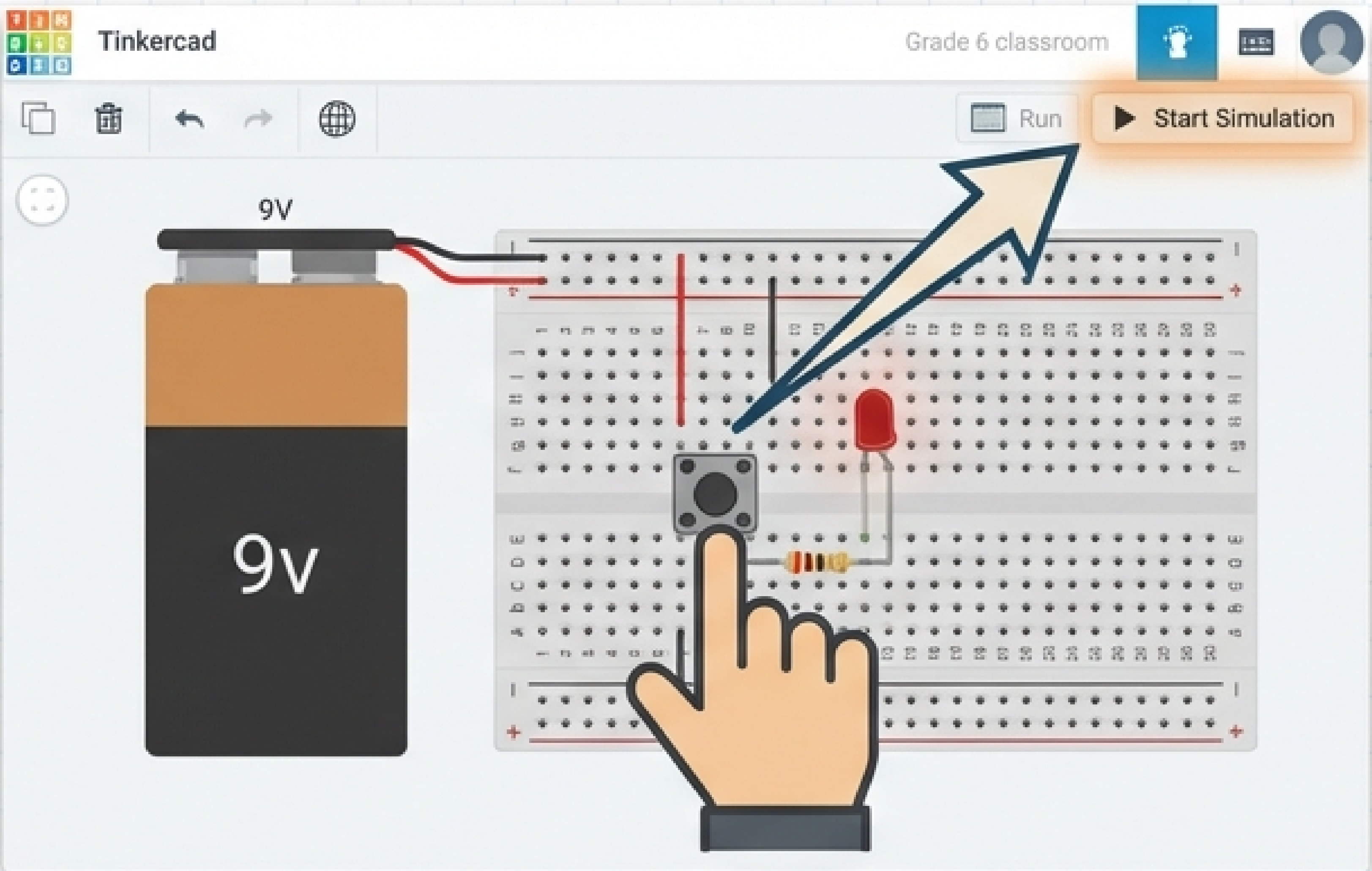


Pro Tip

Hover over the LED legs to see which is Positive (Anode) and Negative (Cathode)!

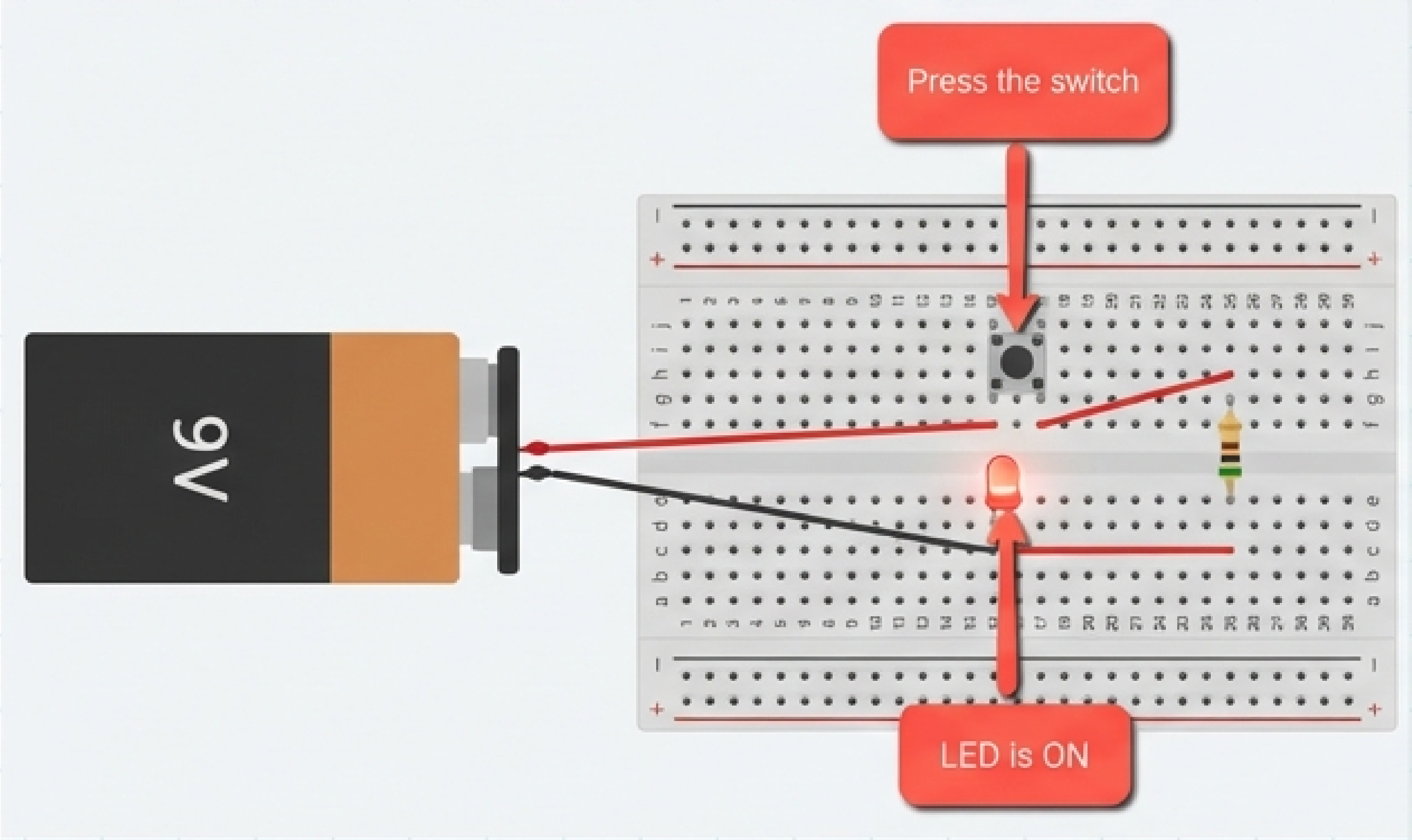
Simulation Time!

1. Click **Start Simulation** (Top Right).
2. Click and **Hold** the Pushbutton.
3. RESULT: The LED turns **ON!**
4. Release: The LED turns **OFF**.



Success! You have created a functional Closed Circuit.

The Math Behind the Light



Battery Voltage (V) = **9V**

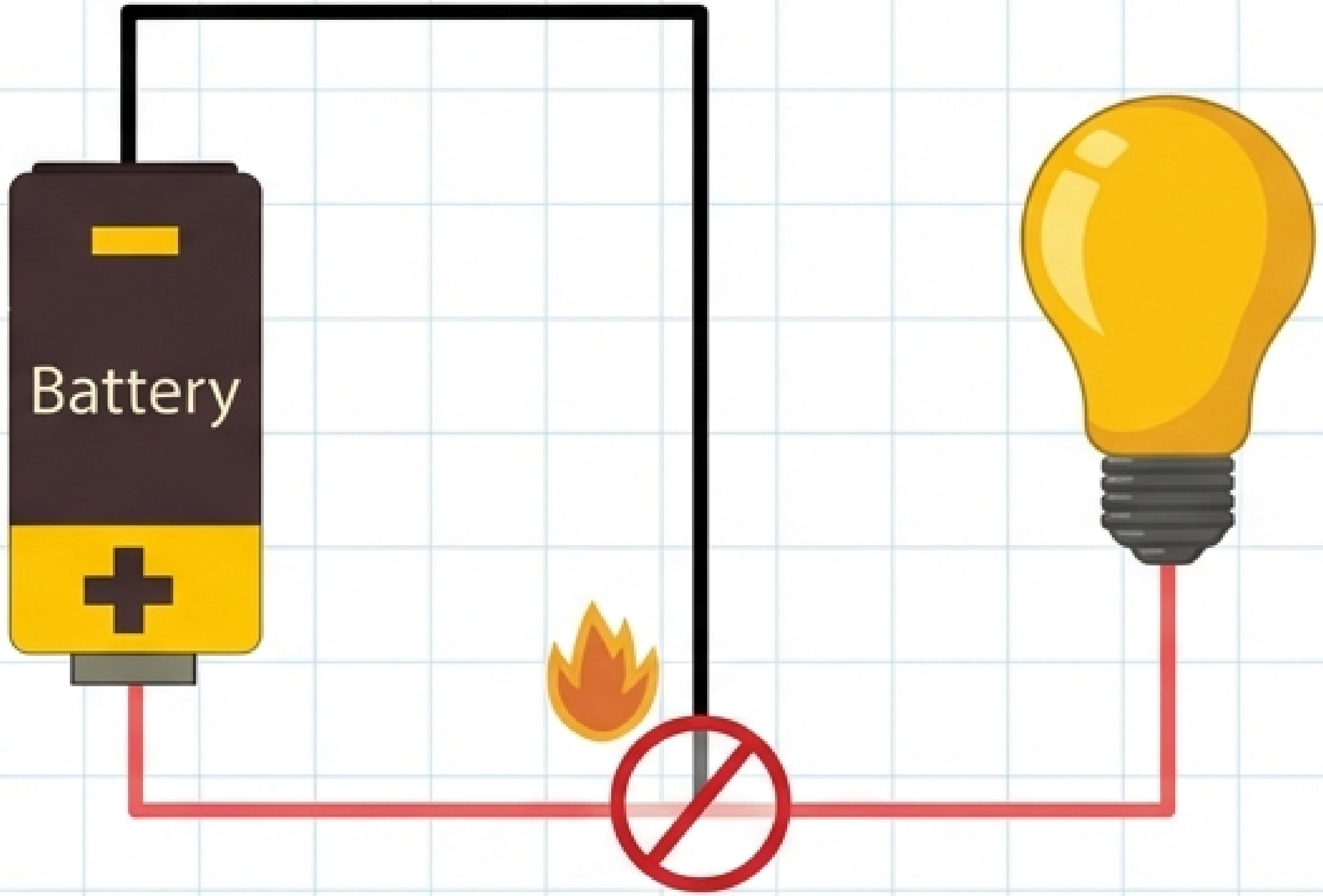
Resistor Value (R) = **500 Ω**

Ohm's Law: $I = V / R$

$I = 9 / 500 = \boxed{0.018\ A}$

An LED needs about **0.020 A** to be safe. Our math proves the resistor is perfect!

Danger Zone: The Short Circuit



What happens if we remove the resistor (0 Ω)?
- The current flows too fast.
- Heat builds up immediately.
- Sparks, Fire, or Broken Components!

Final Safety Rule: NEVER connect **Positive** directly to **Negative** without a **Load** (like a resistor).